$$\frac{x : \tau \in \Gamma}{\Gamma \vdash_{\text{val}} x : \tau}\ \text{T-E-Var} \qquad \frac{\Gamma, x : \tau_1 \vdash e : \tau_2 \mid \varepsilon}{\Gamma \vdash_{\text{val}} \lambda^{\varepsilon} x : \tau_1 . e : \tau_1 \to^{\varepsilon} \tau_2}\ \text{T-E-Abs}$$

$$\frac{\Gamma, x : \tau_1 \vdash_{\text{val}} v : \tau}{\Gamma \vdash_{\text{val}} \mu x : \tau_1 . v : \tau}\ \text{T-E-Fix} \qquad \frac{\Gamma \vdash_{\text{val}} v : \tau}{\Gamma \vdash v : \tau \mid \varepsilon}\ \text{T-E-Val}$$

$$\frac{k \neq \texttt{lab} \qquad \Gamma \vdash_{\text{val}} v : \tau}{\Gamma \vdash_{\text{val}} \Lambda \alpha^k . v : \forall \alpha^k . \tau}\ \text{T-E-TAbs} \qquad \frac{\Gamma \vdash e_1 : \tau_1 \to^{\varepsilon} \tau_2 \mid \varepsilon \quad \Gamma \vdash e_2 : \tau_1 \mid \varepsilon}{\Gamma \vdash e_1\ e_2 : \tau \mid \varepsilon}\ \text{T-E-App}$$

$$\frac{\vdash_{\text{wf}} \tau : k \qquad \Gamma \vdash e : \forall \alpha^k . \tau_1 \mid \varepsilon}{\Gamma \vdash e[\tau] : \tau_1[\alpha := \tau] \mid \varepsilon}\ \text{T-E-Tapp}$$

$$\frac{\mathsf{op} : \forall \alpha . \tau_1 \to \tau_2 \in \Sigma(l) \qquad \alpha \notin \text{ftv}(\Gamma)}{\Gamma \vdash_{\text{val}} \texttt{perform}^{\varepsilon}\ \mathsf{op}\ \tau : \tau_1[\alpha := \tau] \to^{\langle l \mid \varepsilon \rangle} \tau_2[\alpha := \tau]}\ \text{T-E-Perform}$$

$$\frac{\mathsf{op}_i : \forall \alpha . \tau_1 \to \tau_2 \in \Sigma(l) \qquad \alpha \notin \text{ftv}(\varepsilon, \tau) \quad \Gamma \vdash_{\text{val}} f_i : \forall \alpha . \tau_1 \to^{\varepsilon} ((\tau_2 \to^{\varepsilon} \tau) \to^{\varepsilon} \tau)}{\Gamma \vdash_{\text{ops}} d : \{\mathsf{op}_1 \to f_1, ..., \mathsf{op}_n \to f_n\} : \tau \mid l \mid \varepsilon}\ \text{T-E-Ops-Deep}$$

$$\frac{\mathsf{op}_i : \forall \alpha . \tau_1 \to \tau_2 \in \Sigma(l) \qquad \alpha \notin \text{ftv}(\varepsilon, \tau) \quad \Gamma \vdash_{\text{val}} f_i : \forall \alpha . \tau_1 \to^{\varepsilon} \left(\left(\tau_2 \to^{\langle l,\ \varepsilon \rangle} \tau\right) \to^{\varepsilon} \tau\right)}{\Gamma \vdash_{\text{ops}} s : \{\mathsf{op}_1 \to f_1, ..., \mathsf{op}_n \to f_n\} : \tau \mid l \mid \varepsilon}\ \text{T-E-Ops-Shallow}$$

$$\frac{\Gamma \vdash_{\text{ops}} m : h : \tau \mid l \mid \varepsilon}{\Gamma \vdash_{\text{val}} \texttt{handler}^{\varepsilon}_{m}(h) : \left(\mathsf{unit} \to^{\langle l \mid \varepsilon \rangle} \tau\right) \to^{\varepsilon} \tau}\ \text{T-E-Handler}$$

$$\frac{\Gamma \vdash_{\text{ops}} m : h : \tau \mid l \mid \varepsilon \qquad \Gamma \vdash e : \tau \mid \langle l \mid \varepsilon \rangle}{\Gamma \vdash \texttt{handle}^{\varepsilon}_{m}(h, e) : \tau \mid \varepsilon}\ \text{T-E-Handle}$$

Figure 1: Type Rules for **Effects**